

Pseudo-code for Transient Brokerage Simulation

1 State, Notation, and Matching Function

Blue notation, equations, and algorithm lines are diagnostics-only measures. They are recorded for $\mathcal{M}$ or downstream figures, exploration scripts, and analyses; they do not enter transition, choice, learning, matching, or client-demand acquisition rules. Dark-red notation and algorithm lines are specific to the optional client-demand acquisition model.

For any matrix Z , $Z_{\cdot\ell}$ denotes column ℓ and $Z_{\cdot L}$ denotes the submatrix with ordered column-index set L . For a vector z , z_L denotes the corresponding subvector.

Table 1: Cross-routine symbols and dimensions.

Symbol	Dimension/domain	Definition
<i>Core</i>		
N	scalar	Number of agents, with agent set $[N] = \{1, \dots, N\}$.
ϑ	parameter vector	Simulation parameter vector.
seed	integer	Random seed.
S^t	state tuple	Mutable simulation state at end of period t .
T	integer	Number of simulation periods.
$\mathcal{M} = [m^1; \dots; m^T]$	$\mathbb{R}^{T \times m}$	Measure table; row $m^t = \mathcal{M}_t \in \mathbb{R}^m$.
<i>Parameters</i>		
d	scalar	Agent type dimension.
k_G	scalar	Watts-Strogatz graph degree.
ω	scalar	Satisfaction/confidence EWMA weight.
E_{init}	scalar	Initial training steps.
η_r	scalar	MLP learning rate.
<i>Agent types, networks, and output</i>		
$G^t = (V, \mathcal{E}_G^t)$	graph	Undirected graph, $V = [N] \cup \{N + 1\}$; node $N + 1$ is the broker.
$X^t = [x_1^t, \dots, x_N^t]$	$\mathbb{R}^{d \times N}$	Agent type matrix; $x_i^t = X_{\cdot i}^t \in \mathbb{R}^d$.
γ	$[0, 1] \rightarrow \mathbb{R}^d$	Agent type curve for initial and entrant draws.
q_{ij}^t	scalar	Realized output for match (i, j) .
<i>True matching function</i>		
Q	scalar	Output offset.
ρ	scalar	Quality-interaction weight.
δ	scalar	Regime-gain amplitude.
σ_ε	scalar	Output-noise standard deviation.
c	$\mathbb{R}^d$	Ideal agent type.
A	$\mathbb{R}^{d \times d}$	Symmetric positive definite interaction matrix.
B	$\mathbb{R}^{d \times d}$	Symmetric regime operator.
<i>Calibration and costs</i>		
$\bar{q}_{\text{cal}}$	scalar	Calibration mean.
r	scalar	Reservation output.

Symbol	Dimension/domain	Definition
ϕ	scalar	Standard broker fee.
c_s	scalar	Self-search cost.
<i>Learning and predictions</i>		
$\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$	$\Xi_i^t \in \mathbb{R}^{d \times n_i^t}, y_i^t \in \mathbb{R}^{n_i^t}$	Agent i history; observation ℓ is partner agent type $\Xi_{i,\ell}^t$ and output $y_{i\ell}^t$.
$\mathcal{H}_b^t = (\Xi_b^t, y_b^t, n_b^t)$	$\Xi_b^t \in \mathbb{R}^{2d \times n_b^t}, y_b^t \in \mathbb{R}^{n_b^t}$	Broker history; observation ℓ has ordered-pair input $\Xi_{b,\ell}^t = [x_u; x_v]$ and output $y_{b\ell}^t$.
Θ_i^t	NN parameters	Agent i MLP weights.
Θ_b^t	NN parameters	Broker MLP weights.
$v_b^t(i, j)$	scalar	Symmetric broker prediction for unordered pair $\{i, j\}$.
$w_{j \leftarrow i}^t$	scalar	Receiver acceptance score.
$\bar{q}_i^t$	vector	Agent i 's partner-indexed empirical means.
C_i^t	vector	Agent i 's partner-indexed counts.
<i>Period state and market objects</i>		
s_i^t	$\mathbb{R}^2$	Satisfaction scores for self and broker channels.
χ_i^t	binary	Indicator that agent i has ever used the broker.
M_i^t	set	Agent i current-period counterparties.
z_i^t	channel	Channel chosen by demander i .
d_i^t	integer	Current-period demand.
D^t	subset of $[N]$	Current broker clients, $\{i : d_i^t > 0, z_i^t = \text{broker}\}$.
$\mathcal{A}_b^t$	subset of $[N]$	Broker access set, $\text{Roster}^t \cup D^t$.
Roster^t	subset of $[N]$	Standing roster.
rep_b^t	scalar	Broker reputation.
D_+^t	subset of $[N]$	Broker clients with positive residual offer quota.
$\mathcal{D}^t$	list	Demand records (i, z_i^t, d_i^t) with $d_i^t > 0$.
$\mathcal{R}^t$	list	Accepted records: standard $(i, j, z, 0, q, \hat{q}, o_1, o_2)$; principal $(i, j, z, 1, q, \hat{q}, p, o_1, o_2)$.
<i>Capture readiness</i>		
κ_b^t	scalar	EWMA broker exposure MAE.
$n_{\kappa,b}^t$	integer	Cumulative broker exposure-error count.
CapReady_b^t	binary	Principal-mode readiness gate.

The mutable simulation state at the end of period t is

$$S^t = \left(X^t, G^t, \gamma, \{\mathcal{H}_i^t, \Theta_i^t, \bar{q}_i^t, C_i^t, s_i^t, \chi_i^t, M_i^t\}_{i=1}^N, \mathcal{H}_b^t, \Theta_b^t, \text{Roster}^t, D^t, \text{rep}_b^t, \kappa_b^t, n_{\kappa,b}^t \right).$$

The matching function is

$$f(x_i, x_j) = \rho \frac{x_i^\top c + x_j^\top c}{2} + (1 - \rho) g(x_i, x_j) x_i^\top A x_j, \quad g(x_i, x_j) = 1 + \delta \text{sign}(x_i^\top B x_j).$$

2 Top-Level Simulation

Algorithm 1 Transient Brokerage Simulation

Require: Parameters ϑ and seed.

Ensure: Final state S^T and **measure table** $\mathcal{M} \in \mathbb{R}^{T \times m}$.

- 1: $S^0 \leftarrow \text{INITIALIZE}(\vartheta, \text{seed})$
- 2: **for** $t = 1, \dots, T$ **do**

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3:    $(S^t, \mathcal{R}^t) \leftarrow \text{PERIODUPDATE}(S^{t-1}, t, \vartheta)$ 
4:    $\mathcal{M}_t \leftarrow \text{COLLECTMEASURES}(S^t, \mathcal{R}^t, \vartheta)$ 
5: return  $(S^T, \mathcal{M})$ 

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3 Training

For any one-hidden-layer predictor with input dimension p_{in} and hidden width h ,

$$\text{NN}_{\Theta}(\xi) = w_2^\top \text{ReLU}(W_1 \xi + b_1) + b_2, \quad \Theta = (W_1, b_1, w_2, b_2),$$

where $\xi \in \mathbb{R}^{p_{\text{in}}}$, $W_1 \in \mathbb{R}^{h \times p_{\text{in}}}$, $b_1, w_2 \in \mathbb{R}^h$, and $b_2 \in \mathbb{R}$. Agent networks use $(p_{\text{in}}, h) = (d, h_a)$; the broker network uses $(p_{\text{in}}, h) = (2d, h_b)$. For broker symmetry augmentation, let

$$P_d = \begin{bmatrix} 0_{d \times d} & I_d \\ I_d & 0_{d \times d} \end{bmatrix}, \quad P_d[a; b] = [b; a] \quad (a, b \in \mathbb{R}^d).$$

Local variables for training

Symbol	Dimension/domain	Definition
p_{in}, h	integers	Predictor input dimension and hidden width.
ξ	$\mathbb{R}^{p_{\text{in}}}$	Predictor input vector.
Θ	NN parameters	Generic MLP parameters (W_1, b_1, w_2, b_2) .
W_1, b_1, w_2, b_2	matrix, vectors, scalar	One-hidden-layer MLP weights and biases.
h_a, h_b	integers	Agent and broker hidden widths.
$\Xi_{\text{tr}}, y_{\text{tr}}$	$\mathbb{R}^{p_{\text{in}} \times n}, \mathbb{R}^n$	Generic training inputs and targets.
$\mathcal{L}$	scalar	Full-batch MSE objective.
e	integer	Gradient-step index.
W	integer	Training-window length.
n_i, n_b	integers	Current values of history lengths n_i^t and n_b^t .
L_i, L_b	ordered index sets	Training-window observation indices.
Δ_i, Δ_b	integers	New observations since last training call.
E_i, E_b	integers	Agent and broker gradient-step counts.
$\Xi_{i,\text{tr}}, y_{i,\text{tr}}$	$\mathbb{R}^{d \times L_i }, \mathbb{R}^{ L_i }$	Agent i training view of $\mathcal{H}_i^t$.
$\Xi_{b,\text{tr}}, y_{b,\text{tr}}$	$\mathbb{R}^{2d \times 2 L_b }, \mathbb{R}^{2 L_b }$	Symmetry-augmented broker training view of $\mathcal{H}_b^t$.

Algorithm 2 TrainNN

Require: Network parameters Θ , inputs $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}} \times n}$, targets $y_{\text{tr}} \in \mathbb{R}^n$, steps E , learning rate η_{lr} .

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1: for  $e = 1, \dots, E$  do
2:    $\mathcal{L}(\Theta) \leftarrow n^{-1} \sum_{\ell=1}^n (\text{NN}_{\Theta}(\Xi_{\text{tr}, \ell}) - y_{\text{tr}, \ell})^2$ 
3:    $\Theta \leftarrow \Theta - \eta_{\text{lr}} \nabla_{\Theta} \mathcal{L}(\Theta)$ 
4: return  $\Theta$ 

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Within period-level routines, state variables without a superscript refer to their current-period values.

Algorithm 3 TrainPredictors

Require: State S , period t , parameters $(E_{\text{init}}, \eta_{\text{lr}})$.

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1:  $W \leftarrow 500$ 
2: for  $i = 1, \dots, N$  in parallel do
3:    $n_i \leftarrow n_i^t, \Delta_i \leftarrow$  observations added to  $\mathcal{H}_i$  since its last training call.
4:   if  $n_i > 0, \Delta_i > 0$ , and  $i \bmod 2 = t \bmod 2$  then
5:      $L_i \leftarrow$  ordered indices of the most recent  $\min(W, n_i)$  observations in  $\mathcal{H}_i$ .
6:      $\Xi_{i,\text{tr}} \leftarrow (\Xi_i^t)_{L_i}, y_{i,\text{tr}} \leftarrow (y_i^t)_{L_i}$ .
7:      $E_i \leftarrow \max\{50, \lceil E_{\text{init}} \Delta_i / n_i \rceil\}$ .
8:      $\Theta_i \leftarrow \text{TRAINNN}(\Theta_i, \Xi_{i,\text{tr}}, y_{i,\text{tr}}, E_i, \eta_{\text{lr}}); \Delta_i \leftarrow 0$ .
9:    $n_b \leftarrow n_b^t, \Delta_b \leftarrow$  broker observations since last training.
10:  if  $n_b > 0$  and  $\Delta_b > 0$  then
11:     $L_b \leftarrow$  ordered indices of the most recent  $\min(W, n_b)$  observations in  $\mathcal{H}_b$ .
12:     $\Xi_{b,\text{tr}} \leftarrow [(\Xi_b^t)_{L_b}, \mathbf{P}_d(\Xi_b^t)_{L_b}]$ .
13:     $y_{b,\text{tr}} \leftarrow ((y_b^t)_{L_b}, (y_b^t)_{L_b})$ .
14:     $E_b \leftarrow \max\{50, \lceil E_{\text{init}} \Delta_b / n_b \rceil\}$ .
15:     $\Theta_b \leftarrow \text{TRAINNN}(\Theta_b, \Xi_{b,\text{tr}}, y_{b,\text{tr}}, E_b, \eta_{\text{lr}}); \Delta_b \leftarrow 0$ .

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4 Initialization

Calibration draws random pairs (i_ℓ, j_ℓ) and sets

$$\bar{q}_{\text{cal}} = \frac{1}{10,000} \sum_{\ell=1}^{10,000} \{Q + f(x_{i_\ell}, x_{j_\ell})\}, \quad r = 0.60\bar{q}_{\text{cal}}, \quad \phi = c_s = \lambda_c(\bar{q}_{\text{cal}} - r).$$

The capture-readiness gate is

$$\text{CapReady}_b^t = \begin{cases} 1, & n_{\kappa,b}^t \geq n_{\min} \quad \text{and} \quad \kappa_b^t / (\bar{q}_{\text{cal}} - r) \leq \kappa_{\max}, \\ 0, & \text{otherwise,} \end{cases} \quad \text{CapReady}_b^0 = 0.$$

Local variables for Initialize

Symbol	Dimension/domain	Definition
λ_c	scalar	Cost rate used to set ϕ and c_s .
(i_ℓ, j_ℓ)	pair in $[N]^2$	Random calibration pair.
$n_{\min}, \kappa_{\max}$	scalars	Principal-mode confidence gate.
d_γ	integer	Number of active curve dimensions.
p_{rewire}	scalar	Watts-Strogatz rewiring probability.
ν_ℓ	integer	Frequency for active curve coordinate ℓ .
ψ_ℓ	scalar	Phase for active curve coordinate ℓ .
A_0	$\mathbb{R}^{d \times d}$	Raw matrix used to construct A .
Σ_x	$\mathbb{R}^{d \times d}$	Empirical agent type second moment.
B_0	$\mathbb{R}^{d \times d}$	Raw zero-trace symmetric regime matrix.
B_{raw}	$\mathbb{R}^{d \times d}$	Orthogonalized pre-normalization regime matrix.
$\Xi_{b,\text{seed}}, y_{b,\text{seed}}$	$\mathbb{R}^{2d \times 2n_b^0}, \mathbb{R}^{2n_b^0}$	Symmetry-augmented broker seed training batch.

Local variables for InitializeAgent

Symbol	Dimension/domain	Definition
σ_x	scalar	Agent type noise scale.
u_i	scalar	Agent i curve position.
ϵ_i	$\mathbb{R}^d$	Agent type noise for agent i .
π_{ij}^{ent}	scalar	Entrant attachment probability.
$\mathcal{V}_i^{\text{ent}}$	subset of $[N] \setminus \{i\}$	Entrant attachment set.

Algorithm 4 InitializeAgent

Require: State S , agent slot i , period counter t , parameters ϑ .

Ensure: Agent slot i is initialized in S .

- 1: Draw $u_i \sim \text{Unif}[0, 1]$ and $\epsilon_i \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$.
- 2: $x_i^t \leftarrow (\gamma(u_i) + \epsilon_i) / \|\gamma(u_i) + \epsilon_i\|_2$.
- 3: Reset $\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$ to zero observations, $\bar{q}_{ji}^t$, C_{ji}^t , χ_i^t , and M_i^t ; draw fresh agent MLP weights Θ_i^t with output bias Q .
- 4: **if** $t = 0$ **then**
- 5: $s_i^t \leftarrow (0, 0)$.
- 6: **else**
- 7: Remove i from Roster t , D^t , and M_j^t for all j ; delete all incident edges in G^t ; set $\bar{q}_{ji}^t, C_{ji}^t \leftarrow 0$ for all $j \neq i$.
- 8: Sample $\mathcal{V}_i^{\text{ent}} \subset [N] \setminus \{i\}$ without replacement, $|\mathcal{V}_i^{\text{ent}}| = \min(\lfloor k_G/2 \rfloor, N - 1)$, with

$$\pi_{ij}^{\text{ent}} \propto \exp\{-\|x_i^t - x_j^t\|_2^2\}.$$

- 9: Add edges (i, j) to G^t for all $j \in \mathcal{V}_i^{\text{ent}}$.
 - 10: $s_{i,\text{self}}^t \leftarrow \begin{cases} |\mathcal{V}_i^{\text{ent}}|^{-1} \sum_{j \in \mathcal{V}_i^{\text{ent}}} s_{j,\text{self}}^t, & \mathcal{V}_i^{\text{ent}} \neq \emptyset, \\ 0, & \mathcal{V}_i^{\text{ent}} = \emptyset. \end{cases}$
 - 11: $s_{i,\text{broker}}^t \leftarrow \text{rep}_b^t$.
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Algorithm 5 Initialize

Require: Parameters ϑ and seed.

Ensure: Initial state S^0 .

- 1: Draw curve frequencies $\nu_\ell \sim \text{Unif}\{1, \dots, 5\}$ and phases $\psi_\ell \sim \text{Unif}[0, 2\pi)$ for $\ell = 1, \dots, d_\gamma$, defining γ .
- 2: Allocate empty agent and broker containers in S^0 and store γ .
- 3: **for** $i = 1, \dots, N$ **do**
- 4: INITIALIZEAGENT($S^0, i, 0, \vartheta$).
- 5: Draw ideal agent type c as a noisy curve point.
- 6: Draw $A_0 \in \mathbb{R}^{d \times d}$ iid standard normal and set $A \leftarrow A_0^\top A_0 \cdot d / \text{tr}(A_0^\top A_0)$.
- 7: $\Sigma_x \leftarrow N^{-1} \sum_i x_i^0 (x_i^0)^\top$.
- 8: Draw symmetric zero-trace B_0 and set

$$B_{\text{raw}} \leftarrow B_0 - \frac{\text{tr}(\Sigma_x B_0 \Sigma_x A)}{\text{tr}(\Sigma_x A \Sigma_x A)} A, \quad B \leftarrow B_{\text{raw}} / \|B_{\text{raw}}\|_F.$$

- 9: Calibrate $(\bar{q}_{\text{cal}}, r, \phi, c_s)$.

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10: Build  $G^0$  as a Watts-Strogatz graph on  $[N]$  with degree  $k_G$  and rewiring probability  $p_{\text{rewire}}$ ; add
    isolated broker node  $N + 1$ .
11: Initialize  $\mathcal{H}_b^0$ ,  $\Theta_b^0$ , and  $\text{rep}_b^0$ .
12: Initialize capture gate:  $\kappa_b^0 \leftarrow 0$ ,  $n_{\kappa,b}^0 \leftarrow 0$ , and  $\text{CapReady}_b^0 \leftarrow 0$ .
13: Sample  $\text{Roster}^0 \subset [N]$  uniformly with  $|\text{Roster}^0| = \lceil 0.2N \rceil$ ; add broker edges  $(i, N + 1)$  for all
    roster members.
14: for each unordered non-broker edge  $\{i, j\} \in \mathcal{E}_G^0$  do
15:   Draw  $q_{ij}$  once; append  $(x_j^0, q_{ij})$  to  $\mathcal{H}_i^0$  and  $(x_i^0, q_{ij})$  to  $\mathcal{H}_j^0$ ; update  $\bar{q}_{ij}^0, C_{ij}^0, \bar{q}_{ji}^0, C_{ji}^0$ .
16: Let  $\mathcal{B}_{\text{seed}} \leftarrow \{\{i, j\} \in \mathcal{E}_G^0 : i, j \in \text{Roster}^0\}$ ; shuffle and keep at most 100 edges.
17: for each  $\{i, j\} \in \mathcal{B}_{\text{seed}}$  do
18:   Append column  $[x_i^0; x_j^0]$  to  $\Xi_b^0$ ; append the existing edge outcome  $q_{ij}$  to  $y_b^0$ ; increment  $n_b^0$ .
19: if  $n_b^0 > 0$  then
20:   Set  $\text{rep}_b^0$  to mean broker seed output.
21: else
22:   Set  $\text{rep}_b^0 \leftarrow 0$ .
23: Set  $s_{i,\text{self}}^0$  to mean own seed output and  $s_{i,\text{broker}}^0 \leftarrow \text{rep}_b^0$ .
24: for each  $i$  with  $n_i^0 > 0$  do
25:    $\Theta_i^0 \leftarrow \text{TRAINNN}(\Theta_i^0, \Xi_i^0, y_i^0, E_{\text{init}}, \eta_{\text{lr}})$ .
26:  $\Xi_{b,\text{seed}} \leftarrow [\Xi_b^0, P_d \Xi_b^0]$ ,  $y_{b,\text{seed}} \leftarrow (y_b^0, y_b^0)$ .
27:  $\Theta_b^0 \leftarrow \text{TRAINNN}(\Theta_b^0, \Xi_{b,\text{seed}}, y_{b,\text{seed}}, E_{\text{init}}, \eta_{\text{lr}})$ .
28: return  $S^0$ .

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5 Channel Choice

Local variables for ChooseChannel

Symbol	Dimension/domain	Definition
U_{self}	scalar	Self-channel score.
U_{broker}	scalar	Broker-channel score.

Algorithm 6 ChooseChannel

Require: Agent i state (s_i, χ_i) , broker reputation rep_b , RNG.

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1:  $U_{\text{self}} \leftarrow s_{i,\text{self}}$ .
2:  $U_{\text{broker}} \leftarrow \chi_i s_{i,\text{broker}} + (1 - \chi_i) \text{rep}_b$ .
3: if  $U_{\text{self}} > U_{\text{broker}}$  then
4:   return self.
5: if  $U_{\text{broker}} > U_{\text{self}}$  then
6:   return broker.
7: return self w.p. 1/2, else broker  $\triangleright U_{\text{self}} = U_{\text{broker}}$ .

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6 One Period

Local variables for PeriodUpdate

Symbol	Dimension/domain	Definition
K	integer	Active-demand bound.
p_{demand}	scalar	Per-position demand probability.
p_{roster}	scalar	Roster-churn probability.
η_{exit}	scalar	Agent exit probability.
Δ_{net}	integer	Network-measure interval.

Algorithm 7 PeriodUpdate

Require: Previous state S^{t-1} , period t , parameters ϑ .

Ensure: Updated state S^t and accepted records $\mathcal{R}^t$.

- 1: Clear accumulators, set $M_i^t \leftarrow \emptyset$ for all i , and set $D^t \leftarrow \emptyset$.
 - 2: Refresh roster: remove each roster member with probability p_{roster} , then replenish uniformly to size $\lceil 0.2N \rceil$.
 - 3: $\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t\}$.
 - 4: $\mathcal{D}^t \leftarrow \emptyset$.
 - 5: **for** $i = 1, \dots, N$ **do**
 - 6: Draw $d_i^t \sim \text{Binomial}(K, p_{\text{demand}})$.
 - 7: **if** $d_i^t > 0$ **then**
 - 8: $z_i^t \leftarrow \text{CHOOSECHANNEL}(i, \text{rep}_b^{t-1})$.
 - 9: Append (i, z_i^t, d_i^t) to $\mathcal{D}^t$.
 - 10: **if** $z_i^t = \text{broker}$ **then**
 - 11: Add i to D^t .
 - 12: $\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t \cup D^t\}$.
 - 13: $\text{TRAINPREDICTORS}(S, t, \vartheta)$.
 - 14: $\mathcal{R}^t \leftarrow \text{OFFERMARKET}(S, \mathcal{D}^t, \vartheta)$.
 - 15: $\text{UPDATESATISFACTION}(S, \mathcal{D}^t, \mathcal{R}^t)$.
 - 16: **if** $D^t \neq \emptyset$ **then**
 - 17: $\text{rep}_b^t \leftarrow |D^t|^{-1} \sum_{i \in D^t} s_{i, \text{broker}}^t$.
 - 18: **else**
 - 19: $\text{rep}_b^t \leftarrow \text{rep}_b^{t-1}$.
 - 20: $\text{UPDATEBROKERCONFIDENCE}(S, \mathcal{R}^t)$.
 - 21: **if** $t \bmod \Delta_{\text{net}} = 0$ **then**
 - 22: Recompute broker betweenness, Burt constraint, and effective size on G^t .
 - 23: **for** $i = 1, \dots, N$ **do**
 - 24: **if** agent i exits with probability η_{exit} **then**
 - 25: $\text{INITIALIZEAGENT}(S^t, i, t, \vartheta)$.
 - 26: **return** $(S^t, \mathcal{R}^t)$.
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7 Offer Market

Local variables for OfferMarket

Symbol	Dimension/domain	Definition
$\hat{q}_i^t$	$\mathbb{R}^d \rightarrow \mathbb{R}$	Agent predictor.

Local variables for OfferMarket (continued)

Symbol	Dimension/domain	Definition
$\hat{q}_b^t$	$\mathbb{R}^{2d} \rightarrow \mathbb{R}$	Broker predictor.
$\mathcal{N}_G^t(i)$	subset of $[N]$	Non-broker graph neighbors of i .
U^t	subset of $[N]$	Period stranger pool.
$n_{\text{strangers}}$	integer	Period stranger-pool size.
$v_{i \rightarrow j}^{\text{self}, t}$	scalar	Self-offer score.
Ω_i	integer	Residual offer quota for demander i .
Ω	$\mathbb{Z}_{\geq 0}^N$	Residual quota vector.
O^t	map	Directed offer book.
O_{ij}^t	offer record	Record (i, j, z, v) for offer $i \rightarrow j$.
$\mathcal{P}_O^t$	set	Unordered pairs represented in O^t .
$\mathcal{Z}^t$	subset of $[N]$	Acquired clients removed from standard offers.
$\mathcal{U}_i^{\text{self}}$	subset of $[N]$	Self-search candidate set.
$\mathcal{I}_i^{\text{self}}$	list	Self-search candidates sorted by score.
$\mathcal{P}_b^t$	set	Feasible unordered broker-pair set.
$\mathcal{I}_b$	list	Broker pairs sorted by v_b^t .
o, o', o_{ij}, o_{ji}	offer records	Offer records; fields include $o.z$ and $o.from$.
z^*	channel	Accepted-record channel.
α, β	indices	Ordered endpoints induced by one unordered broker pair.
ι	$\{0, 1\}$	Return value of AddOffer.

The symmetric broker score is

$$v_b^t(i, j) = \frac{1}{2} [\hat{q}_b^t([x_i^t; x_j^t]) + \hat{q}_b^t([x_j^t; x_i^t])].$$

Let $\mathcal{N}_G^t(i) = \{j \in [N] : (i, j) \in \mathcal{E}_G^t\}$ exclude the broker node. The self-search proposal score is defined only for observed graph neighbors and period strangers:

$$v_{i \rightarrow j}^{\text{self}, t} = \begin{cases} \bar{q}_{ij}^t, & j \in \mathcal{N}_G^t(i), C_{ij}^t > 0, \\ \hat{q}_i^t(x_j^t), & j \in U^t \setminus \mathcal{N}_G^t(i), \\ \perp, & \text{otherwise.} \end{cases}$$

Thus direct pair means dominate predictor extrapolations when pair-specific experience exists; $\perp$ denotes no valid self-search score. The receiver-side acceptance score for a one-sided offer is

$$w_{j \leftarrow i}^t = \begin{cases} \bar{q}_{ji}^t, & (i, j) \in \mathcal{E}_G^t, C_{ji}^t > 0, \\ \hat{q}_j^t(x_i^t), & \text{otherwise.} \end{cases}$$

The market creates at most d_i^t outgoing offers from demander i . Receivers have no per-period capacity constraint; acceptance is pairwise, so a receiver may accept multiple offers in the same period. A demanded position can remain unmatched if no acceptable offer is formed.

Algorithm 8 AddOffer

Require: Offer book O^t , unordered-pair set $\mathcal{P}_O^t$, offer (i, j, z, v) .

1: **if** O_{ij}^t is defined **then**

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2:   return 0.
3:  $O_{ij}^t \leftarrow (i, j, z, v)$ .
4: if  $O_{ji}^t$  is undefined then
5:    $\mathcal{P}_O^t \leftarrow \mathcal{P}_O^t \cup \{\{i, j\}\}$ .
6: return 1.

```

Algorithm 9 OfferMarket

Require: State S , demand records $\mathcal{D}^t$, parameters ϑ .

Ensure: Accepted records $\mathcal{R}^t$.

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1: For all  $(i, z_i, d_i) \in \mathcal{D}^t$ , set residual offer quota  $\Omega_i \leftarrow d_i$ .
2:  $O^t \leftarrow \emptyset$ ,  $\mathcal{P}_O^t \leftarrow \emptyset$ ,  $\mathcal{R}^t \leftarrow \emptyset$ .
3: Draw one period-level stranger pool  $U^t \subset [N]$  uniformly without replacement,  $|U^t| = \min(n_{\text{strangers}}, N)$ .
4:  $\mathcal{Z}^t \leftarrow \emptyset$ .
5: if principal mode is enabled then
6:    $\mathcal{Z}^t \leftarrow \text{ACQUIRECLIENTDEMAND}(S, \mathcal{D}^t, \Omega, \mathcal{R}^t, \vartheta)$ .
7: for each  $(i, \text{self}, d_i) \in \mathcal{D}^t$  with  $\Omega_i > 0$  do
8:    $\mathcal{U}_i^{\text{self}} \leftarrow \{j \in \mathcal{N}_G^t(i) : C_{ij}^t > 0\} \cup (U^t \setminus \mathcal{N}_G^t(i))$ .
9:   Remove from  $\mathcal{U}_i^{\text{self}}$  the elements of  $\{i\} \cup M_i^t \cup \mathcal{Z}^t$ .
10:   $\mathcal{I}_i^{\text{self}} \leftarrow \text{argsort}_{j \in \mathcal{U}_i^{\text{self}}: v_{i \rightarrow j}^{\text{self}, t} > r}(-v_{i \rightarrow j}^{\text{self}, t})$ .
11:  for the first  $\min(\Omega_i, |\mathcal{I}_i^{\text{self}}|)$  agents  $j$  in  $\mathcal{I}_i^{\text{self}}$  do
12:     $\text{ADDOFFER}(O^t, \mathcal{P}_O^t, i, j, \text{self}, v_{i \rightarrow j}^{\text{self}, t})$ .
13:  $\mathcal{A}_b^t \leftarrow (\text{Roster}^t \cup D^t) \setminus \mathcal{Z}^t$ .
14:  $D_+^t \leftarrow \{i : (i, \text{broker}, d_i) \in \mathcal{D}^t, i \notin \mathcal{Z}^t, \Omega_i > 0\}$ .
15:  $\mathcal{P}_b^t \leftarrow \{\{i, j\} : i \neq j, (i \in D_+^t, j \in \mathcal{A}_b^t) \text{ or } (j \in D_+^t, i \in \mathcal{A}_b^t)\}$ .
16:  $\mathcal{I}_b \leftarrow \text{argsort}_{\{i, j\} \in \mathcal{P}_b^t: v_b^t(i, j) > r}(-v_b^t(i, j))$ .
17: for each  $\{i, j\} \in \mathcal{I}_b$  do
18:   for each ordered pair  $(\alpha, \beta) \in \{(i, j), (j, i)\}$  do
19:     if  $\alpha \in D_+^t$ ,  $\beta \in \mathcal{A}_b^t$ , and  $\Omega_\alpha > 0$  then
20:        $\iota \leftarrow \text{ADDOFFER}(O^t, \mathcal{P}_O^t, \alpha, \beta, \text{broker}, v_b^t(i, j))$ .
21:        $\Omega_\alpha \leftarrow \Omega_\alpha - \iota$ .
22: for each  $\{i, j\} \in \mathcal{P}_O^t$  do
23:    $o_{ij} \leftarrow O_{ij}^t$  if defined, else  $\emptyset$ ;  $o_{ji} \leftarrow O_{ji}^t$  if defined, else  $\emptyset$ .
24:   if  $o_{ij} \neq \emptyset$  and  $o_{ji} \neq \emptyset$  then
25:      $\text{FINALIZESTANDARD}(S, \mathcal{R}^t, o_{ij}, o_{ji})$ .
26:   else if  $o_{ij} \neq \emptyset$  and  $w_{j \leftarrow i}^t > r$  then
27:      $\text{FINALIZESTANDARD}(S, \mathcal{R}^t, o_{ij}, \emptyset)$ .
28:   else if  $o_{ji} \neq \emptyset$  and  $w_{i \leftarrow j}^t > r$  then
29:      $\text{FINALIZESTANDARD}(S, \mathcal{R}^t, o_{ji}, \emptyset)$ .
30: return  $\mathcal{R}^t$ .

```

Algorithm 10 FinalizeStandard

Require: State S , accepted-record list $\mathcal{R}^t$, offer $o = (i, j, z, v)$, optional reverse offer o' .

```

1: if  $j \in M_i^t$  then
2:   return .
3: Draw  $q_{ij}^t = Q + f(x_i^t, x_j^t) + \varepsilon$ ,  $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ .

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4: Append columns  $x_j^t$  to  $\Xi_i$  and  $x_i^t$  to  $\Xi_j$ ; append  $q_{ij}^t$  to  $y_i$  and  $y_j$ ; increment  $n_i$  and  $n_j$ .
5: Update  $\bar{q}_{ij}, C_{ij}$  and  $\bar{q}_{ji}, C_{ji}$ .
6: Add edge  $(i, j)$  to  $G^t$ ; add  $j$  to  $M_i^t$  and  $i$  to  $M_j^t$ .
7: if  $z = \text{broker}$  or  $(o' \neq \emptyset \text{ and } o'.z = \text{broker})$  then
8:   Append column  $[x_i^t; x_j^t]$  to  $\Xi_b$ ; append  $q_{ij}^t$  to  $y_b$ ; increment  $n_b$ .
9:    $z^* \leftarrow \text{broker}$ .
10: else
11:    $z^* \leftarrow \text{self}$ .
12: Append accepted record  $(i, j, z^*, 0, q_{ij}^t, v, o, o')$  to  $\mathcal{R}^t$ .

```

8 Optional Client-Demand Acquisition

Local variables for `AcquireClientDemand`

Symbol	Dimension/domain	Definition
p_i	scalar	Acquisition price per acquired position.
$\mathcal{J}_i$	subset of $[N]$	Planned recipient set.
Γ_i	scalar	Expected net acquisition gain.
$\mathcal{I}_{\text{acq}}$	list	Feasible client acquisitions sorted by Γ_i .

Algorithm 11 `AcquireClientDemand`

Require: State S , demand records $\mathcal{D}^t$, residual quotas Ω , accepted records $\mathcal{R}^t$.

Ensure: Acquired clients $\mathcal{Z}^t$; principal records appended to $\mathcal{R}^t$.

```

1: if  $\text{CapReady}_b^t = 0$  then
2:   return  $\emptyset$ .
3:  $D_+^t \leftarrow \{(i, \text{broker}, d_i) \in \mathcal{D}^t, \Omega_i > 0\}$ ,  $\mathcal{A}_b^t \leftarrow \text{Roster}^t \cup D^t$ .
4: Score  $\mathcal{P}_b^t = \{(i, j) : i \in D_+^t, j \in \mathcal{A}_b^t, i \neq j\}$  by  $v_b^t(i, j)$ .
5: for  $i \in D_+^t$  do
6:    $p_i \leftarrow (n_i^t)^{-1} \sum_{\ell=1}^{n_i^t} y_{i\ell}^t$  if  $n_i^t > 0$ , else  $p_i \leftarrow \bar{q}_{\text{cal}}$ .
7:    $\mathcal{J}_i \leftarrow \text{top } \Omega_i \text{ distinct } j \in \mathcal{A}_b^t \setminus \{i\} \text{ with } v_b^t(i, j) > r$ .
8:    $\Gamma_i \leftarrow \sum_{j \in \mathcal{J}_i} v_b^t(i, j) - \Omega_i p_i - \Omega_i \phi$ .
9:  $\mathcal{I}_{\text{acq}} \leftarrow \text{argsort}_{i: |\mathcal{J}_i| = \Omega_i, p_i \geq r, \Gamma_i > 0}(-\Gamma_i)$ ;  $\mathcal{Z}^t \leftarrow \emptyset$ .
10: for  $i \in \mathcal{I}_{\text{acq}}$  do
11:   Recompute  $\mathcal{J}_i$  from the sorted pair list, excluding current matches and  $\mathcal{Z}^t$ .
12:   if  $|\mathcal{J}_i| < \Omega_i$  then
13:     continue.
14:   Add  $i$  to  $\mathcal{Z}^t$ ; record principal payment  $\Omega_i p_i$  to  $i$ ; set  $\Omega_i \leftarrow 0$ .
15:   for  $j \in \mathcal{J}_i$  do
16:     Mark  $(i, j)$  as current before receiver evaluation.
17:     if  $w_{j \leftarrow i}^t > r$  then
18:       Draw  $q_{ij}^t$ ; append column  $[x_i^t; x_j^t]$  to  $\Xi_b$ ; append  $q_{ij}^t$  to  $y_b$ ; increment  $n_b$ .
19:       Add principal active matches  $i \leftrightarrow j$ .
20:       Append  $(i, j, \text{broker}, 1, q_{ij}^t, v_b^t(i, j), p_i, \emptyset, \emptyset)$  to  $\mathcal{R}^t$ .
21:     else
22:       Record rejected principal exposure with realized value 0 and prediction  $v_b^t(i, j)$ .

```

23: **return** $\mathcal{Z}^t$.

Principal positions update broker history only when accepted. They do not update agent histories, agent-agent edges, or agent partner means.

9 Satisfaction, Confidence, and Measures

Local variables for satisfaction and confidence

Symbol	Dimension/domain	Definition
Y_i	scalar	Agent i period satisfaction numerator.
$n_{i,b}^{\text{std}}$	integer	Standard broker offer credits for demander i .
$\tilde{q}_i^t$	scalar	Channel-specific satisfaction input.
$P_i^{\text{acq},t}$	scalar	Principal payment to acquired client i , zero absent acquisition.
ζ	accepted record	Iterator over $\mathcal{R}^t$.
$\mathbf{1}\{\cdot\}$	indicator	Equals 1 when its condition is true and 0 otherwise.
$\mathcal{E}_{\text{err},b}^t$	set	Broker exposure-error observations.
e_b^t	scalar	Period mean absolute exposure error.

Algorithm 12 UpdateSatisfaction

Require: State S , demand records $\mathcal{D}^t$, accepted records $\mathcal{R}^t$.

- 1: **for** each $(i, z_i, d_i) \in \mathcal{D}^t$ **do**
- 2: $Y_i \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset, o.\text{from} = i, \zeta.\pi = 0\} \zeta.q + P_i^{\text{acq},t}$.
- 3: $n_{i,b}^{\text{std}} \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset, o.\text{from} = i, o.z = \text{broker}, \zeta.\pi = 0\}$.
- 4: **if** $z_i = \text{self}$ **then**
- 5: $\tilde{q}_i^t \leftarrow Y_i/d_i - c_s$.
- 6: **else**
- 7: $\tilde{q}_i^t \leftarrow (Y_i - \phi n_{i,b}^{\text{std}})/d_i$; $\chi_i^t \leftarrow 1$.
- 8: $s_{i,z_i}^t \leftarrow (1 - \omega)s_{i,z_i}^{t-1} + \omega \tilde{q}_i^t$.

Algorithm 13 UpdateBrokerConfidence

Require: State S , accepted records $\mathcal{R}^t$.

- 1: $\mathcal{E}_{\text{err},b}^t \leftarrow \emptyset$.
- 2: **for** each accepted record $\zeta \in \mathcal{R}^t$ **do**
- 3: **for** each standard offer credit $o \in \{\zeta.o_1, \zeta.o_2\}$ **do**
- 4: **if** $o \neq \emptyset$ and $o.z = \text{broker}$ **then**
- 5: Add $(o.v, \zeta.q)$ to $\mathcal{E}_{\text{err},b}^t$.
- 6: **if** $\zeta.\pi = 1$ **then**
- 7: Add $(\zeta.\hat{q}, \zeta.q)$ to $\mathcal{E}_{\text{err},b}^t$.
- 8: **if** $|\mathcal{E}_{\text{err},b}^t| > 0$ **then**
- 9: $e_b^t \leftarrow |\mathcal{E}_{\text{err},b}^t|^{-1} \sum_{(\hat{q}, q) \in \mathcal{E}_{\text{err},b}^t} |\hat{q} - q|$.
- 10: $\kappa_b^t \leftarrow e_b^t$ if $n_{\kappa,b}^{t-1} = 0$, else $(1 - \omega)\kappa_b^{t-1} + \omega e_b^t$.
- 11: $n_{\kappa,b}^t \leftarrow n_{\kappa,b}^{t-1} + |\mathcal{E}_{\text{err},b}^t|$.

CollectMeasures. $\text{COLLECTMEASURES}(S^t, \mathcal{R}^t, \vartheta)$ is a pure measure-collection function. It records demand, outsourcing, accepted matches, realized output, selected-sample prediction quality, deterministic holdout prediction quality, broker access and assessment counts, satisfaction, reputation, roster size, broker access size, agent-node degree summaries excluding the broker node, cached broker network measures, and principal-mode diagnostics when enabled. The measures themselves do not feed back into S^{t+1} .